

CS 230 : Discrete Computational Structures

Fall Semester, 2023

ASSIGNMENT #7

Due Date: Thursday, October 26

Suggested Reading: Rosen Sections 5.2 - 5.3; Lehman et al. Chapters 5, 6.1 - 6.3

For the problems below, explain your answers and show your reasoning.

1. [8 Pts] Prove that $f_0 - f_1 + f_2 - \cdots - f_{2n-1} + f_{2n} = f_{2n-1} - 1$ where f_i are the Fibonacci numbers.
2. [8 Pts] Consider the following state machine with six states, labeled 0, 1, 2, 3, 4 and 5. The start state is 0. The transitions are $0 \rightarrow 1$, $0 \rightarrow 2$, $1 \rightarrow 3$, $2 \rightarrow 4$, $3 \rightarrow 5$, $4 \rightarrow 5$, and $5 \rightarrow 0$.
Prove that if we take n steps in the state machine we will end up in state 0 if and only if n is divisible by 4. Argue that to prove the statement above by induction, we first have to *strengthen the induction hypothesis*. State the strengthened hypothesis and prove it.
3. [8 Pts] Let $P(n)$ be the statement that n can be written as a sum of distinct powers of 2. For example, $5 = 2^0 + 2^2$ and $10 = 2^1 + 2^3$. Note that $10 = 2^0 + 2^0 + 2^3$ is not a sum of *distinct* powers of 2. Prove that $P(n)$ is true for all positive integers n by strong induction.
4. [8 Pts] A robot wanders around a 2-dimensional grid. He starts out at (0,0) and can take the following steps: (+1,+3), (+1,-1), (-4,0) and (0,+4). Define a state machine for this problem. Then, define a Preserved Invariant and prove that the robot will never get to (1,1).
5. [8 Pts] Show that if a predicate $P(n)$ can be proven true for all positive integers n by strong induction, then it can be proven true also by regular induction, once you strengthen the inductive hypothesis. In other words, Strong Induction isn't really stronger than Regular Induction. *Hint: Given any statement $P(n)$, define a new (stronger) statement $Q(n)$ so that proving $P(n)$ by strong induction is similar to proving $Q(n)$ by regular induction.*

For more practice, you are encouraged to work on other problems in Rosen Sections 5.2 and 5.3 and in LLM Chapter 5, 6.1 - 6.3.